Enhanced Entity-Relationship (EER) Modeling

Session Objectives

- ERR Diagram
- Superclass/Subclass
- Specialization /Generalization
- Disjoint/Overlap
- Total/ Partial
- Hierarchy/Lattice
- Union Types

Session Outcomes

- At the end of this session, participants will be able to
  - Understand EER model concepts
  - Understand Superclass/Subclass
  - Understand Generalization/Specialization
  - Understand Disjoint/Overlap

EER MODEL

- EER stands for **Enhanced ER or Extended ER**
- Current applications such as CAD/CAM, telecommunications, GIS,... have more complex requirements
- Led to the development of semantic data modelling concepts
- ER model can enhanced to include **semantic data model** leading to Enhanced ER [EER] model

EER MODEL

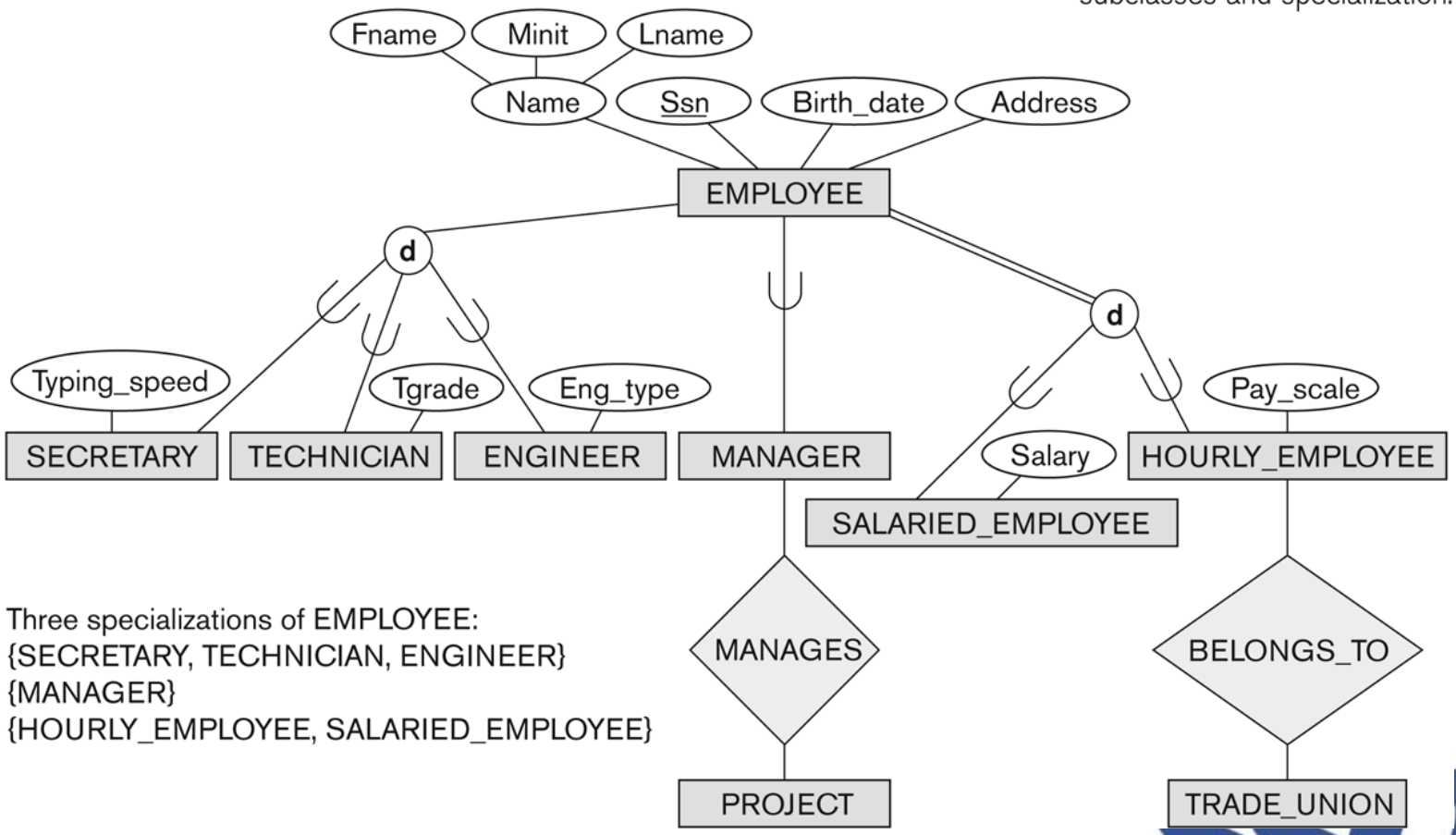
- EER Model Concepts
  - Includes all modeling concepts of basic ER
  - Additional concepts:
    - Subclasses/superclasses
    - Specialization/generalization
    - Attribute and relationship inheritance
- These are fundamental to **conceptual modeling**
- The additional EER concepts are used to model applications more completely and more accurately
- EER includes some object-oriented concepts, such inheritance

Superclass and Subclass

- An entity type may have additional meaningful subgroupings of its entities
  - Example: EMPLOYEE may be further grouped into:
    - SECRETARY, ENGINEER, TECHNICIAN, …
      - Based on the EMPLOYEE's Job
    - MANAGER
      - EMPLOYEEs who are managers
    - SALARIED_EMPLOYEE, HOURLY_EMPLOYEE
      - Based on the EMPLOYEE's method of pay
- EER diagrams extend ER diagrams to represent these additional subgroupings, called *subclasses* or *subtypes*
- The subclass that define a specialization are attached by lines to the circle which is connected to the superclass

Superclass and Subclass

Figure 4.1
EER diagram notation to represent subclasses and specialization.



Three specializations of EMPLOYEE:
{SECRETARY, TECHNICIAN, ENGINEER}
{MANAGER}
{HOURLY_EMPLOYEE, SALARIED_EMPLOYEE}

Subclasses and Superclasses

- Each of these subgroupings is a subset of EMPLOYEE entities
- Each is called a **subclass** of EMPLOYEE
- EMPLOYEE is the **superclass** for each of these subclasses
- These are called superclass/subclass relationships:
  - **EMPLOYEE/SECRETARY**
  - **EMPLOYEE/TECHNICIAN**
  - **EMPLOYEE/MANAGER**

Subclasses and Superclasses

These are also called IS-A relationships

- SECRETARY IS-A EMPLOYEE, TECHNICIAN IS-A EMPLOYEE, ….

Note: An entity that is member of a subclass represents the same real-world entity as some member of the superclass:

- The subclass member is the same entity in a *distinct specific role*
  - A specific **technician** (subclass) is still an **employee** (superclass) and shares its attributes.
- An entity cannot exist in the database merely by being a member of a subclass; it must also be a member of the superclass.
  - You cannot have a "SECRETARY" in the database without that person being an "EMPLOYEE."
- A member of the superclass can be optionally included as a member of any number of its subclasses
  - An employee may not necessarily be a technician or secretary but could be both.

Subclasses and Superclasses

- May have several specializations of the same superclass
- A salaried employee who is also an engineer belongs to the two subclasses:

  ENGINEER,

  SALARIED_EMPLOYEE
- A salaried employee who is also an engineering manager belongs to the three subclasses:

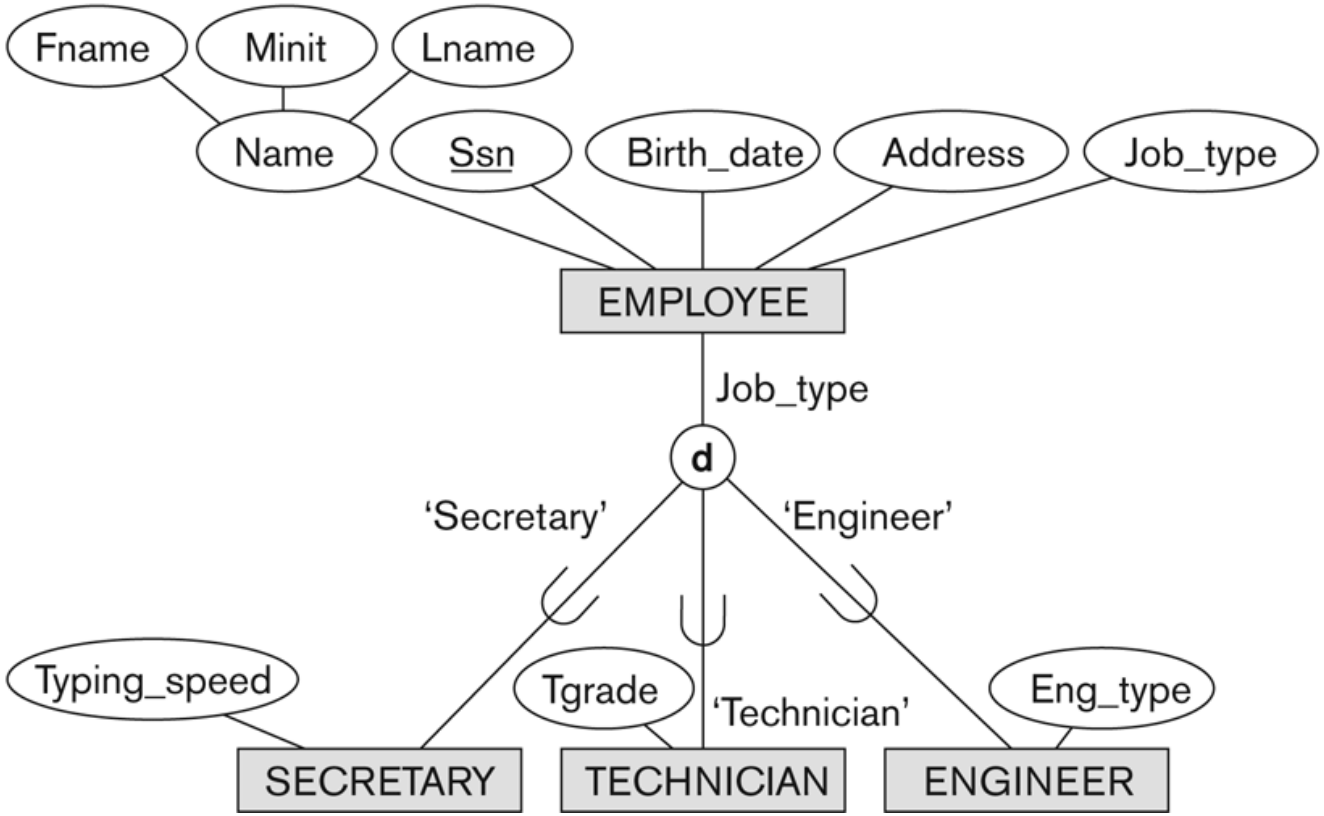
  MANAGER,

  ENGINEER, and

  SALARIED_EMPLOYEE
- SECRETARY (as well as TECHNICIAN and ENGINEER) inherit the attributes Name, SSN, …, from EMPLOYEE
- It is not necessary that every entity in a superclass be a member of some subclass

Representing Specialization in EER Diagrams

Figure 4.4
EER diagram notation for an attribute-defined specialization on Job_type.



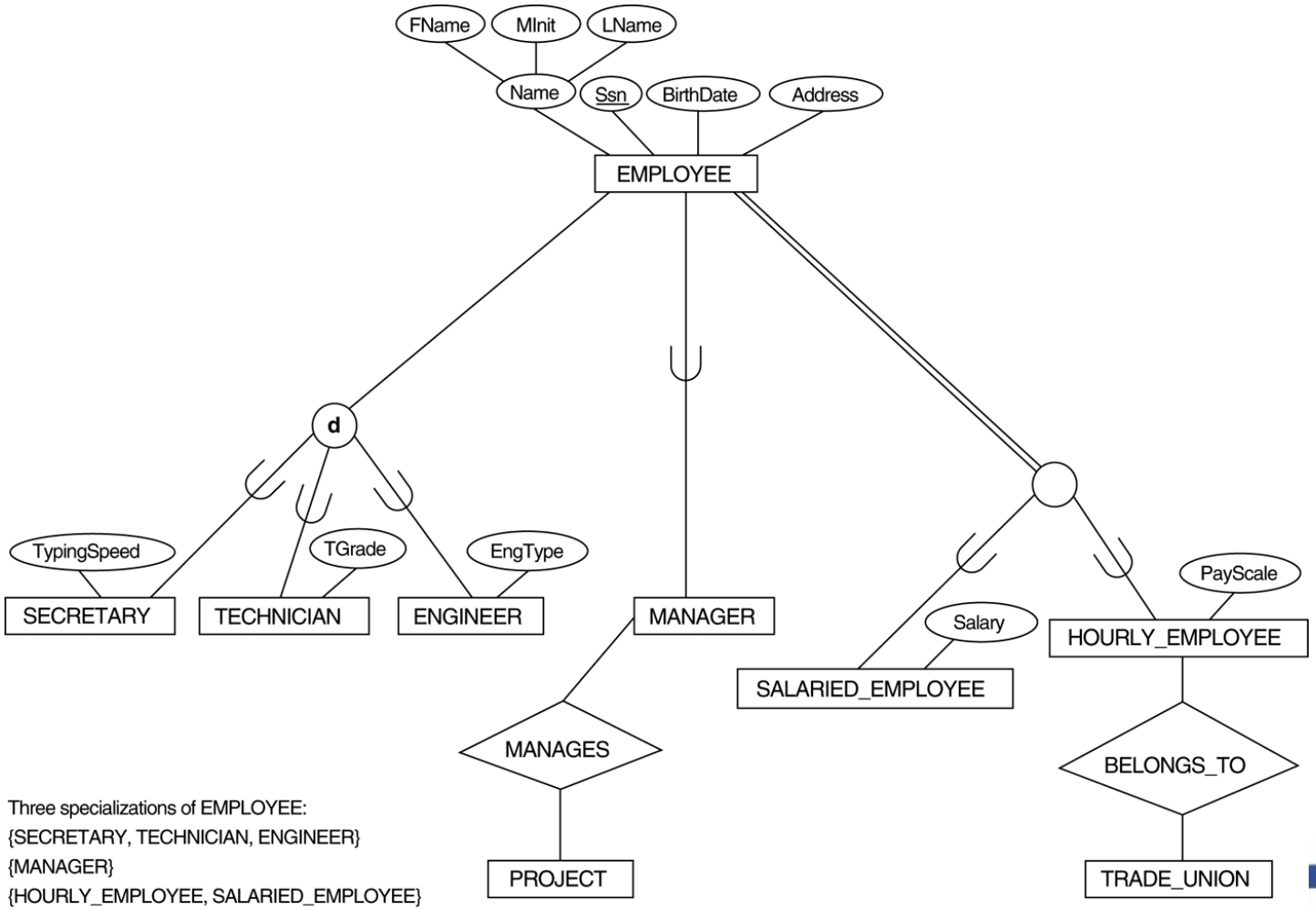
Specialization

- An entity that is member of a subclass inherits
  - **All attributes of the entity as a member of the superclass**
  - **All relationships of the entity as a member of the superclass**
- Specialization is the process of defining a set of subclasses of a superclass
  - based upon some **distinguishing characteristics** of the entities in the superclass
- Example: {SECRETARY, ENGINEER, TECHNICIAN} is a specialization of EMPLOYEE **based upon job type.**
- Another specialization of EMPLOYEE based on method of pay is {SALARIED_EMPLOYEE, HOURLY_EMPLOYEE}.

Specialization

- Attributes of a subclass are called specific or local attributes
  - For example, the attribute TypingSpeed of SECRETARY
- The subclass can also participate in specific relationship types.
  - For example, a relationship BELONGS_TO of HOURLY_EMPLOYEE
- Superclass/subclass relationship resembles a 1:1 relationship at the instance level.

Subclasses and Superclasses - Example



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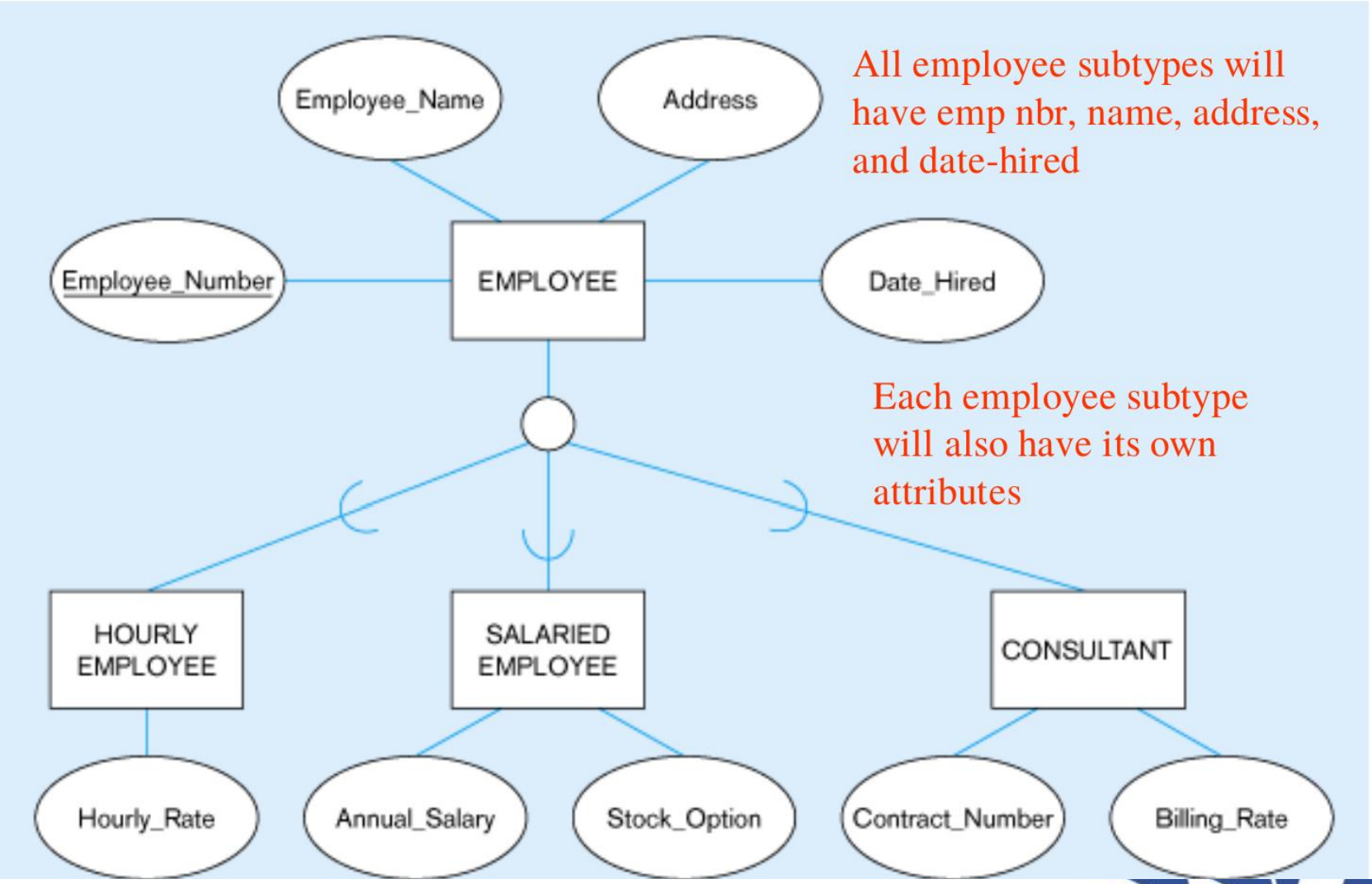
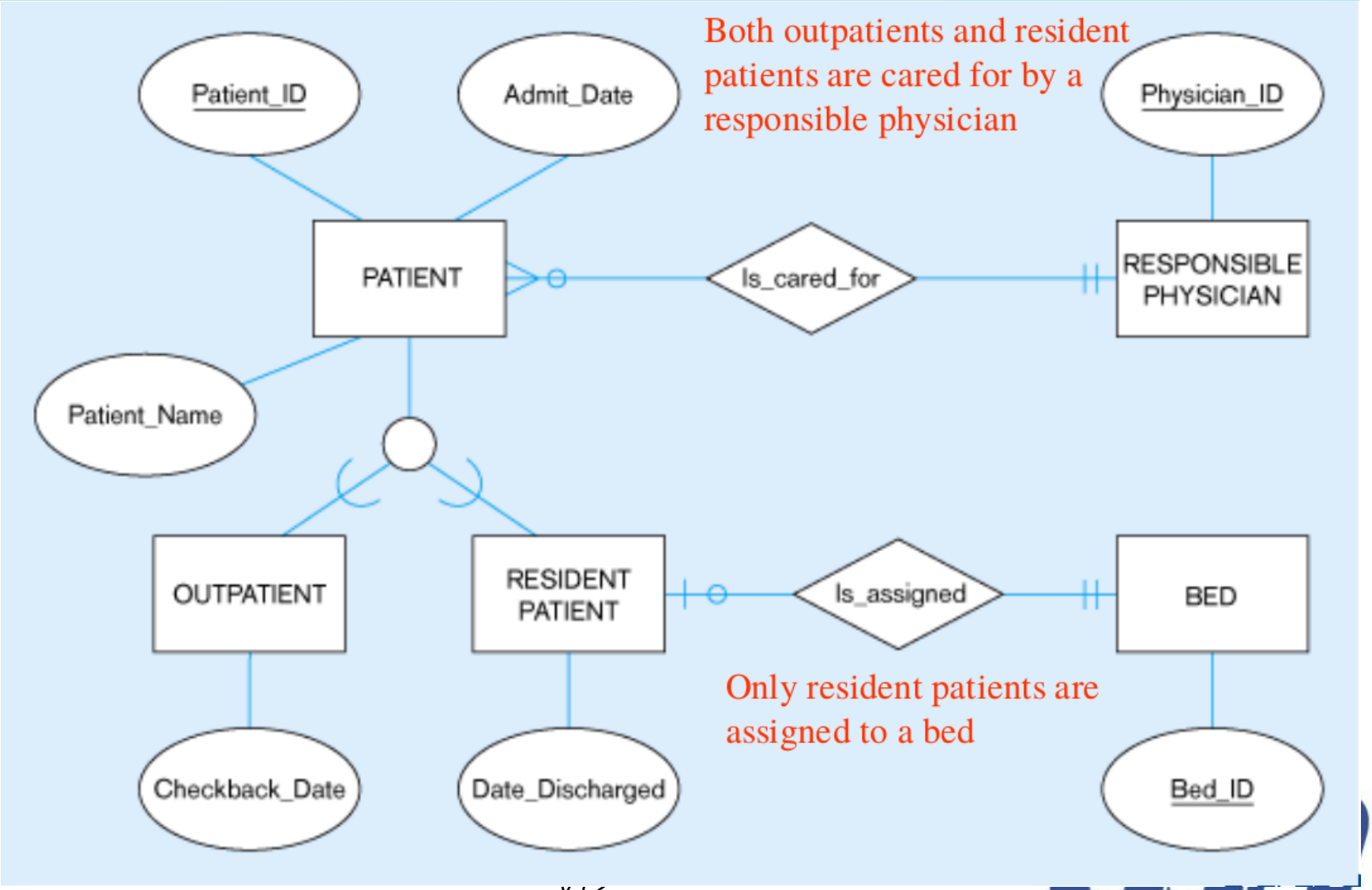
Specialization



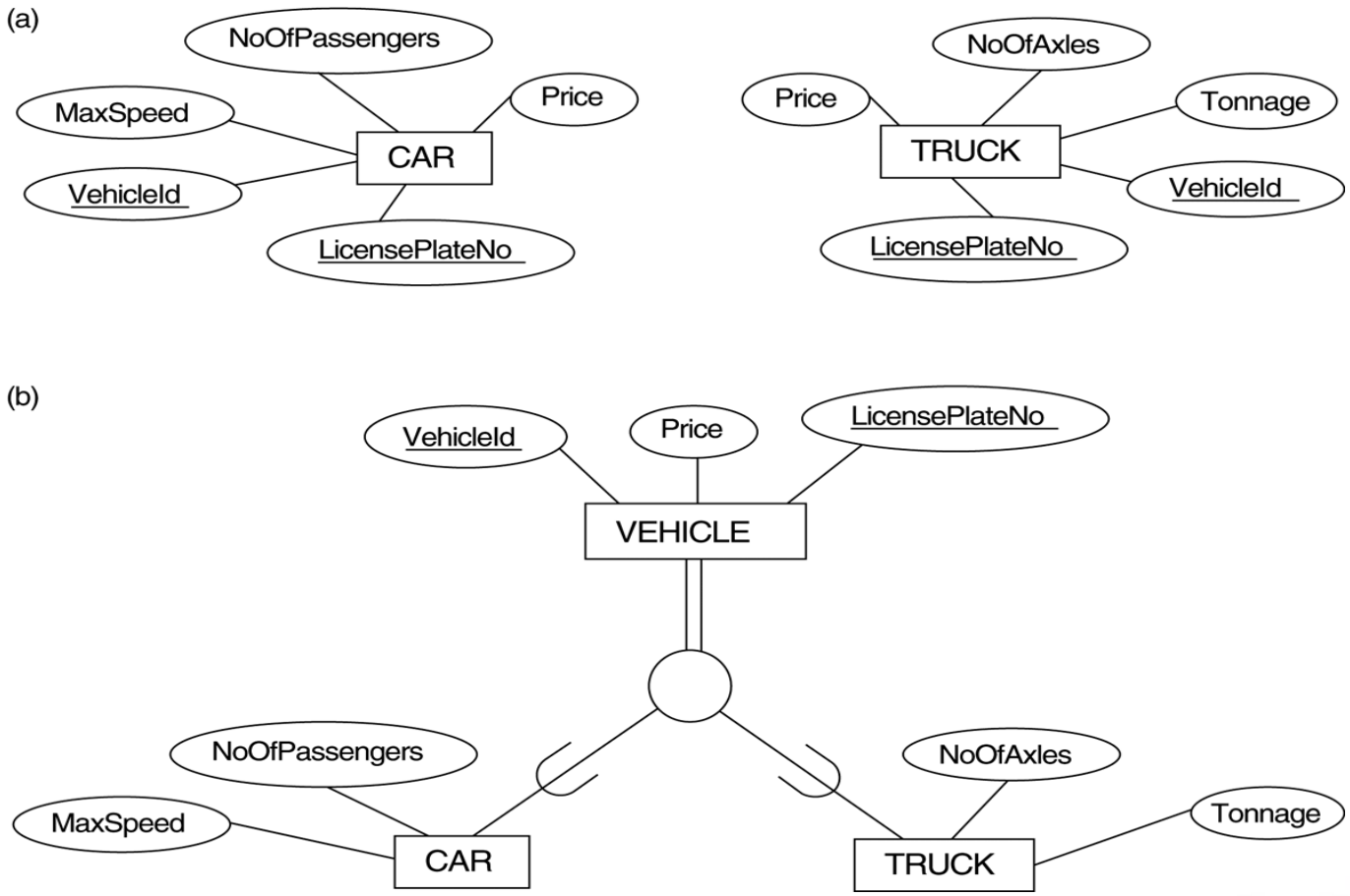
Specialization - Example



Generalization

- Generalization is the reverse of the specialization process
- Several classes with common features are generalized into a superclass
- Example: CAR, TRUCK generalized into VEHICLE;
- Both CAR, TRUCK become subclasses of the superclass VEHICLE.
- Diagrammatic notation are sometimes used to distinguish between generalization and specialization – but it is subjective

Generalization. (a) Two entity types, CAR and TRUCK. (b) Generalizing CAR and TRUCK into the superclass VEHICLE.



Types of Subclass

- **Predicate-Defined (Condition-Defined) Subclass :**
  - Membership in the subclass is determined by a **condition (predicate)** on a **superclass attribute**.
  - Only entities that **satisfy the condition** belong to the subclass.
  - The condition is called the **defining predicate**.
  - In EER diagrams, the predicate is **written next to the line** connecting the subclass to the specialization circle.

Example:

Superclass: EMPLOYEE Attribute: Job_type

Subclass: SECRETARY Predicate: (Job_type = 'Secretary')

- **Attribute-Defined Specialization :**
  - A specialization is called **attribute-defined** if **all subclasses** are defined using conditions on the **same attribute** of the superclass.
  - That attribute is called the **defining attribute**.
  - Entities with the **same value** of the defining attribute belong to the **same subclass**.
  - In EER diagrams, the **defining attribute name** is written next to the arc from the specialization circle to the superclass.

Types of Subclass

Example:
Defining attribute: Job_type
Subclasses: SECRETARY, ENGINEER, MANAGER

- **User-Defined Subclass**
  - No condition is specified for determining subclass membership.
  - Membership is decided **explicitly by the database user**.
  - Entities are added to the subclass **manually**, not automatically.
  - No attribute or predicate controls membership.
  - Useful when subclass membership **cannot be determined algorithmically**.

Constraints on Specialization/Generalization

Two basic constraints can apply to a Specialization/generalization:

– Disjointness Constraint:

– Completeness Constraint:

Constraints on Specialization/Generalization

Disjointness Constraint: Specifies that the subclasses of the specialization must be disjoint:

- an entity can be a member of at most one of the subclasses of the specialization
- Specified by d in EER diagram

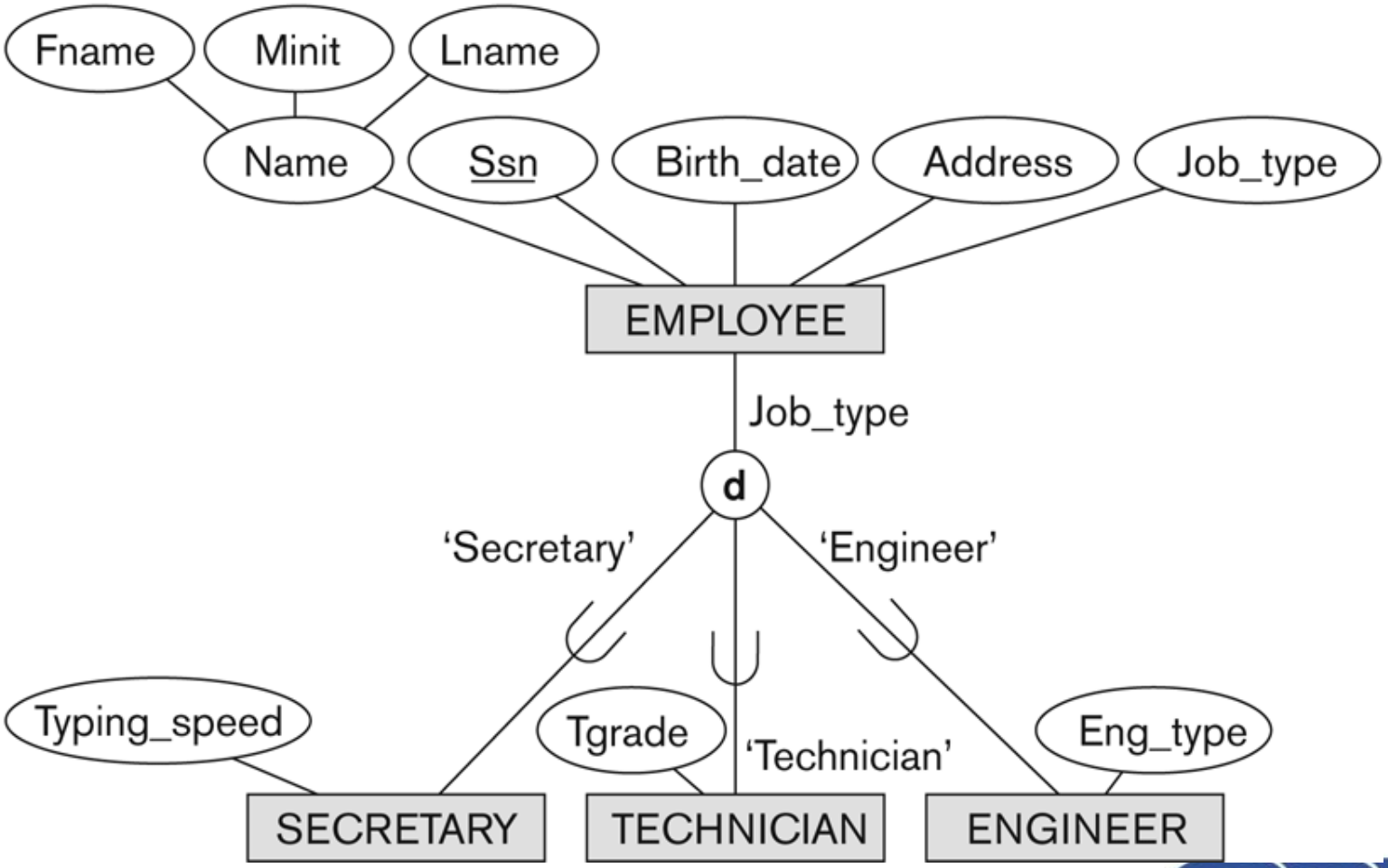
– If not disjoint, specialization is overlapping:

- that is the same entity may be a member of more than one subclass of the specialization
- Specified by o in EER diagram

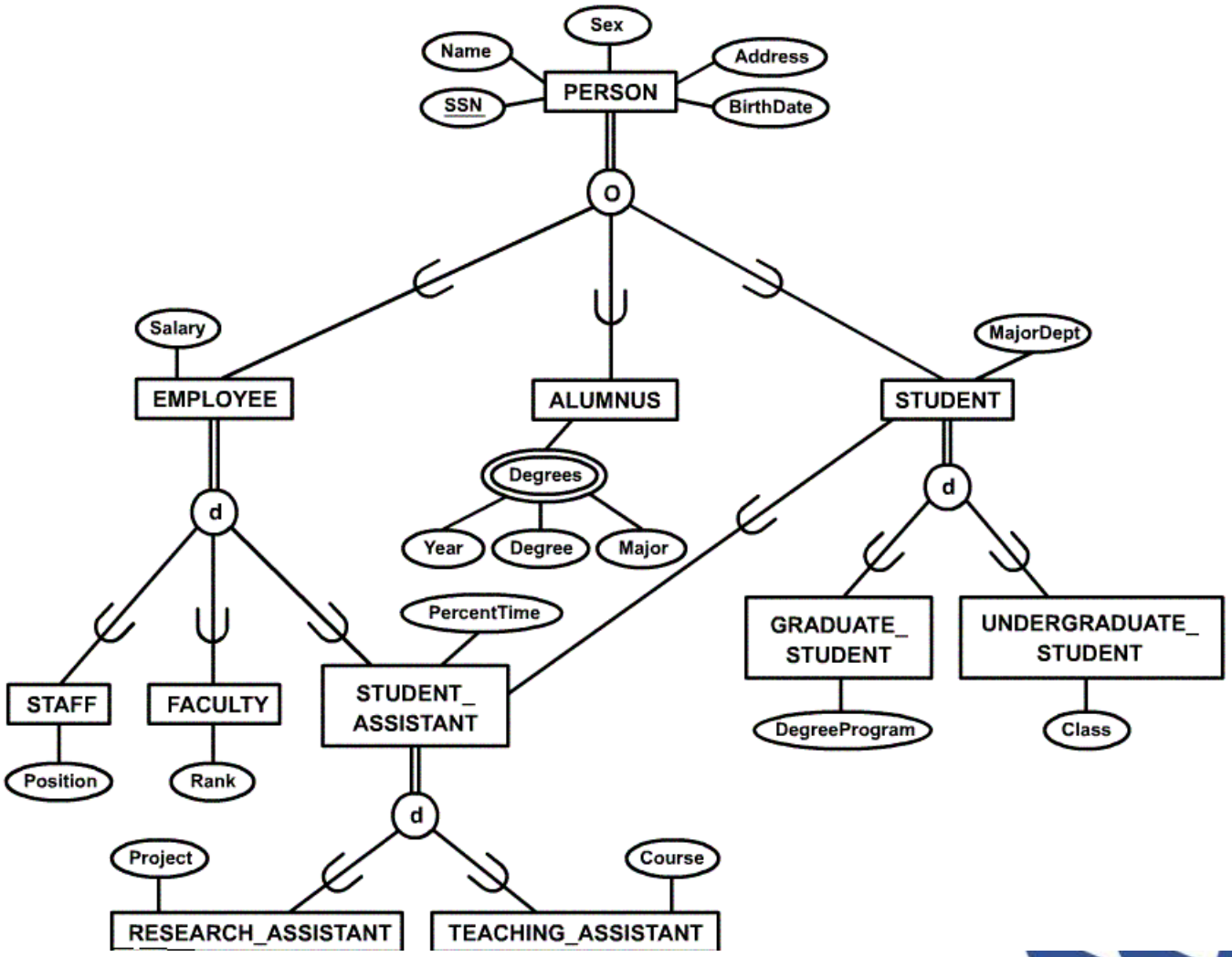
Constraints on Specialization/Generalization

- Completeness Constraint:
  - Total specifies that every entity in the superclass must be a member of some subclass in the specialization/generalization
  - Shown in EER diagrams by a double line
  - Partial allows an entity not to belong to any of the subclasses
  - Shown in EER diagrams by a single line
- Hence, we have four types of specialization/generalization:
  - Disjoint total, Disjoint partial
  - Overlapping total, Overlapping partial
- **Note: Generalization usually is total because the superclass is derived from the subclasses.**

Example of disjoint partial Specialization



Example of overlapping total Specialization



Specialization/Generalization Hierarchies, Lattices & Shared Subclasses

- A subclass may itself have further subclasses specified on it forms a hierarchy or a lattice
- ***Hierarchy*** has a constraint that every subclass has only one superclass (called ***single inheritance***); this is basically a ***tree structure***
- In a ***lattice***, a subclass can be subclass of more than one superclass (called ***multiple inheritance***)

Shared Subclass "Engineering_Manager"

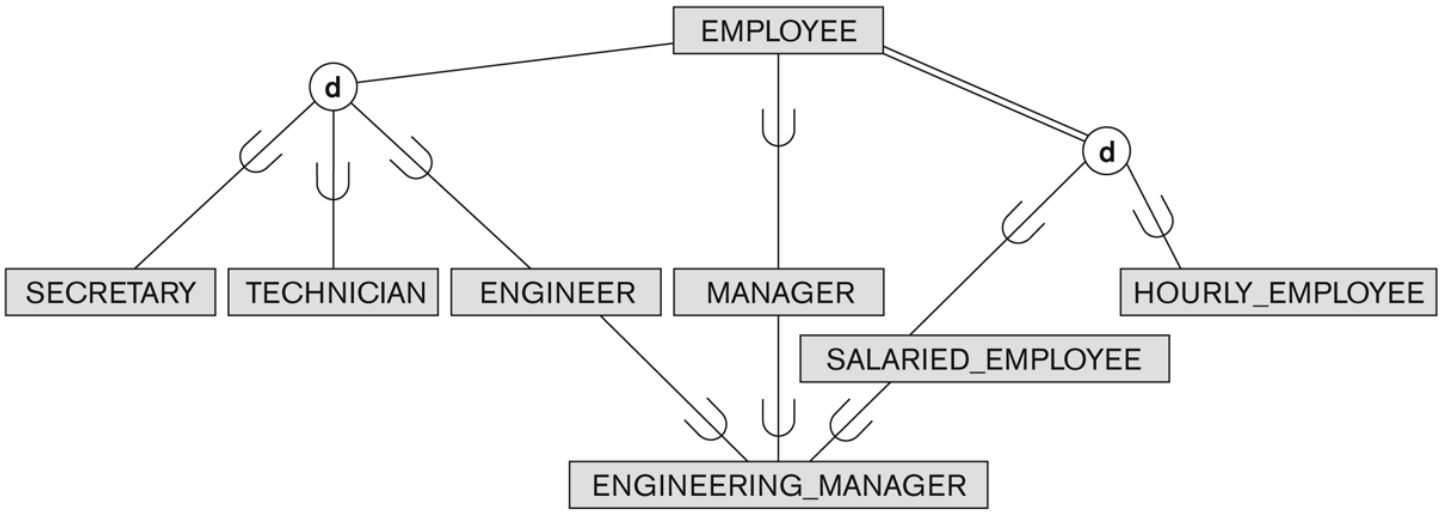


Figure 4.6
A specialization lattice with shared subclass ENGINEERING_MANAGER.

Specialization/Generalization Hierarchies, Lattices & Shared Subclasses

- In a lattice or hierarchy, a subclass inherits attributes not only of its direct superclass, but also of all its predecessor superclasses.

- A subclass with more than one superclass is called a shared subclass (multiple inheritance)

Can have:

- specialization hierarchies or lattices, or
- generalization hierarchies or lattices,
- depending on how they were derived

- We just use specialization (to stand for the end result of either specialization or generalization)

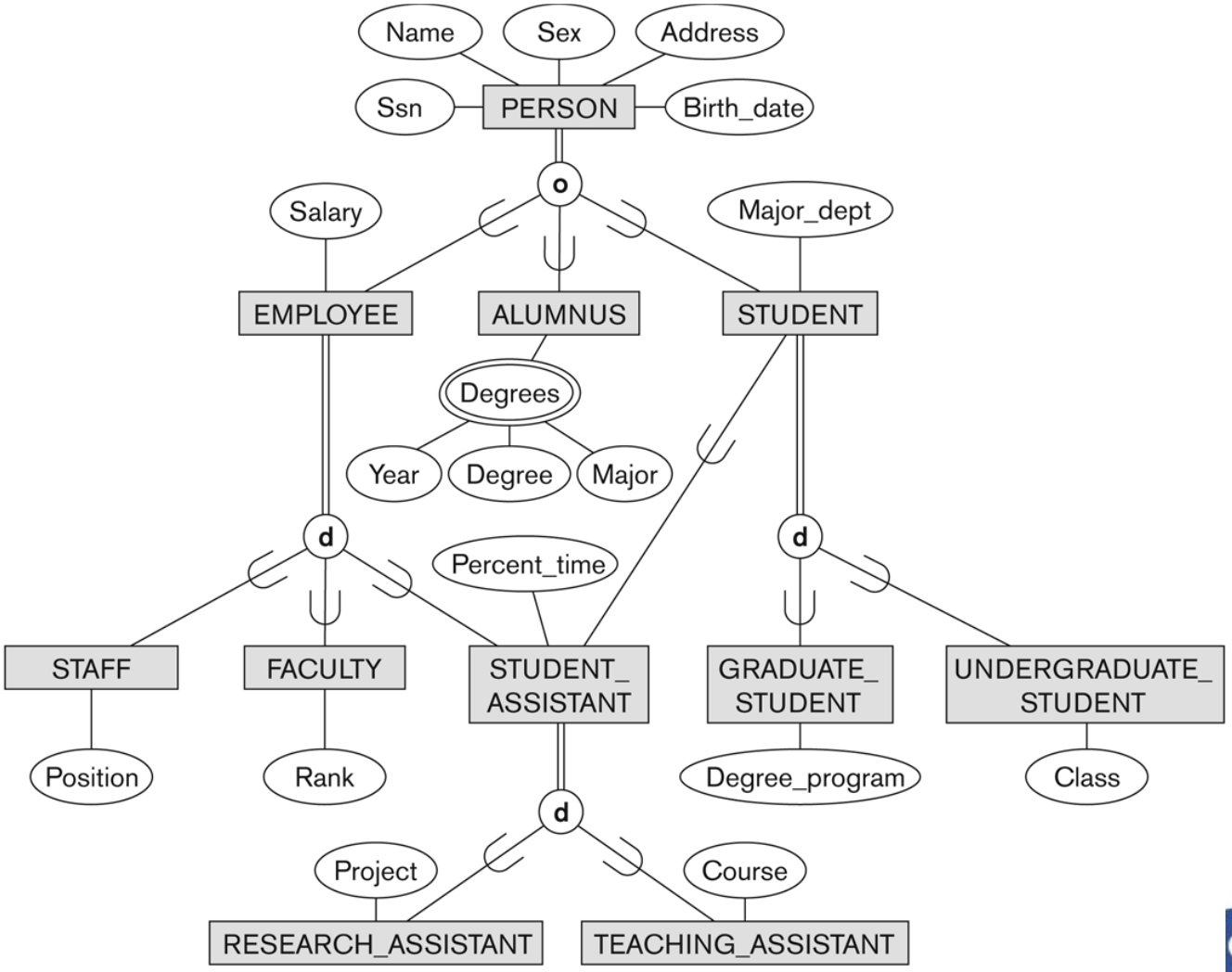
Specialization/Generalization Hierarchies, Lattices & Shared Subclasses

- In specialization, start with an entity type and then define subclasses of the entity type by successive specialization called a **top down conceptual refinement process.**

- In generalization, start with many entity types and generalize those that have common properties
  - Called a **bottom up conceptual synthesis process**

- In practice, a combination of both processes is usually employed

Specialization / Generalization Lattice Example (UNIVERSITY)



Categories (UNION TYPES)

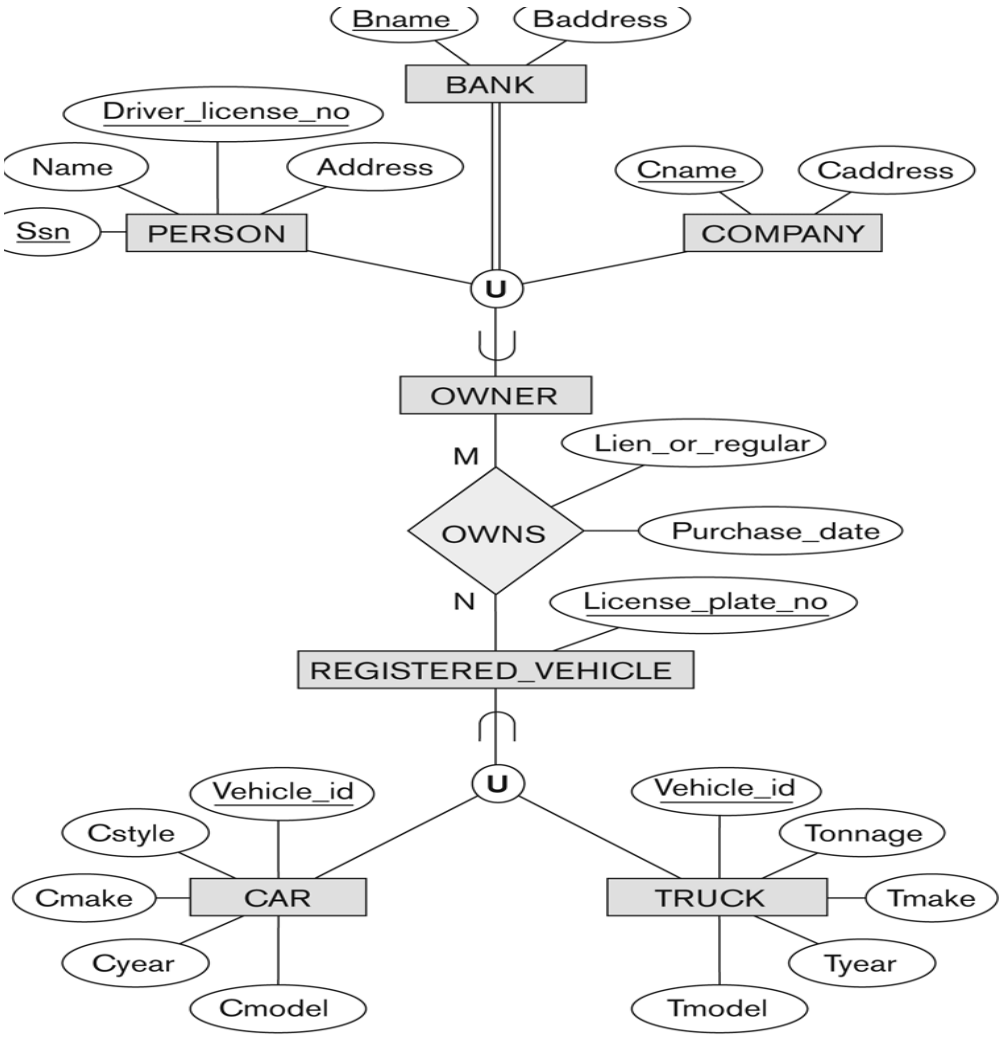
All of the superclass/subclass relationships we have seen thus far have a single superclass

- A shared subclass is a subclass in:
  - more than one distinct superclass/subclass relationships
  - each relationships has a single superclass
  - shared subclass leads to multiple inheritance
- In some cases, we need to model a single superclass/subclass relationship with more than one superclass
  - Superclasses can represent different entity types
  - Such a subclass is called a category or UNION TYPE

Categories (UNION TYPES)

- Example: In a database for vehicle registration, a vehicle owner can be a PERSON, a BANK (holding a lien on a vehicle) or a COMPANY.
- A category (UNION type) called OWNER is created to represent a subset of the union of the three super classes
  - COMPANY, BANK, and PERSON
- A category member must exist in at least one of its superclasses
- Difference from shared subclass, which is a:
  - subset of the intersection of its superclasses
  - shared subclass member must exist in all of its superclasses

Two categories (UNION types): OWNER, REGISTERED_VEHICLE



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types):
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Summary

- ERR Diagram
- Superclass/Subclass
- Specialization /Generalization
- Disjoint/Overlap
- Total/ Partial
- Hierrachy/ Lattices
- Union Type